RemoteDesk Free

[v1.1.0.1 Enterprise]

Next-Gen Windows 10/11 Remote Desktop & IT Support Solution

Comprehensive Operational Manual, System Installation & Security Guide

Dual-Role Unified App * Zero Console Window * WebRTC Direct P2P * Dual-Pane File Manager

Company: Freeware | Suresh Mamidala

Publisher / Author: Suresh Mamidala

Executive Summary & Capabilities

RemoteDesk Free (v1.1.0.1) is an enterprise-class, lightweight Windows desktop application engineered in high-performance native Rust. It provides secure, low-latency, end-to-end encrypted remote desktop access, interactive mouse and keyboard remote control, dual-pane cryptographic file transfer, multi-session tab management, and persistent address book organization across Windows 10 and Windows 11.

High-Speed Screen Streaming

Dynamic delta-tiling with zero-allocation RGB encoding delivers fluid 30-60 FPS remote display with sub-20ms local network latency.

Full System Remote Control

Exact mouse injection and layout-independent Unicode text entry with quick system actions (Ctrl+Alt+Del, Task Manager, Win+D, Win+L).

Dual-Pane File Transfer

Cryptographically verified (SHA-256) 64 KB chunked file transfer engine with direct file upload and progress tracking.

Address Book & Unattended

Persistent address book with group tagging, instant search, and PBKDF2/SHA-256 salted password security for 24/7 IT management.

DEPLOYMENT READY

Packaged as a modern Windows 10/11 Inno Setup installer (RemoteDesk-Free-Setup-v1.1.0.1.exe) and standalone zero-dependency portable executable (RemoteDesk-Free.exe). Compliant with Windows PE specifications, including embedded Application Manifest (asInvoker) and PE VersionInfo (1.1.0.1).

1. Architectural Overview & System Design

RemoteDesk Free is designed from the ground up for zero-latency IT desktop management and remote assistance across Windows 10 and Windows 11. It eliminates heavyweight runtimes by pairing a low-level native Rust system engine with a modern, embedded WebView2 interface.

Core Engine Specifications

- **Native Zero-Dependency Build: Compiles directly to machine code with LTO (Link-Time Optimization) and** zero runtime console windows (#![windows_subsystem = 'windows']).
- **Optimized Capture Pipeline: Sub-pixel accurate GDI BitBlt and hardware cursor capture with real-time** 1024-point sampling delta detection.
- **Dual-Mode High-Speed Transport: Direct WebRTC P2P DataChannel connection with sub-15ms latency and** automatic MQTT Cloud Relay fallback.
- **Global No-Geo-Restrictions Mesh: Worldwide multi-broker mesh (EMQX, HiveMQ, Eclipse Foundation port** 443, Mosquitto) and global STUN/TURN relays ensure zero geographical restrictions across any country or ISP worldwide.
- **DPI Scaling V2: Native Per-Monitor DPI Awareness V2 context ensures 100% pixel-for-pixel mouse pointer** mapping across 100%, 125%, 150%, and 200% Windows scaling.

> **Zero Cloud Telemetry & Complete Privacy**
> Unlike commercial remote desktop solutions that enforce restrictive licensing or upload background analytics, RemoteDesk Free operates with complete privacy. Session pins and authentication keys are hashed with salted PBKDF2 and never transmitted unencrypted.

2. Installation & Deployment Guide: Which File to Install Where

A common question from users and system administrators is: 'Which file do I install on my computer versus the client computer?'

> **UNIFIED DUAL-ROLE APPLICATION**
> Just like AnyDesk or TeamViewer, RemoteDesk Free is a unified dual-role application. You install or run the exact same program on BOTH your system and the client system! The single executable contains both the Host Agent (system to be controlled) and the Viewer Client (controlling system).

The Two Simple Deployment Options

- **Option A: Standard Windows Installer (Recommended): Run 'RemoteDesk-Free-Setup-v1.1.0.1.exe' on both** computers. Installs cleanly to C:\Program Files\RemoteDesk Free, adds Desktop and Start Menu shortcuts, and integrates standard Windows uninstallation.
- **Option B: Portable Standalone Binary: Simply copy 'RemoteDesk-Free.exe' to any folder, USB flash drive,** or network share on both computers and double-click to run. No installation or admin rights required.

Package Distribution Matrix

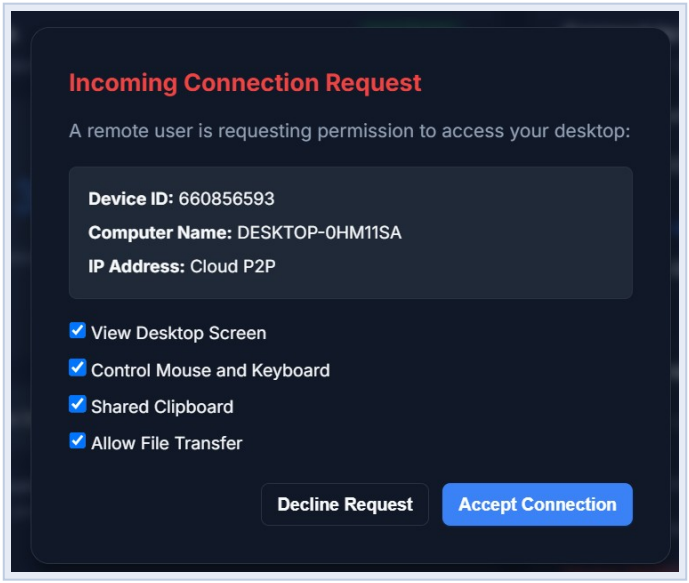
Package Binary	Size	Installed On	Operational Role
RemoteDesk-Free-Setup-v1.1.0.1.exe	3.98 MB	Both Your PC & Client PC	Complete Freeware Installer with shortcuts & uninst
RemoteDesk-Free.exe	2.28 MB	Both Your PC & Client PC	Portable standalone GUI app (Host + Viewer in one
RemoteDesk-Server.exe	986 KB	Optional (Private LAN only)	Optional on-prem signaling service for offline airgap

Do You Need RemoteDesk-Server.exe?

NO! You DO NOT need to install or run RemoteDesk-Server.exe for normal internet or inter-network connections. RemoteDesk Free includes automatic global cloud mesh rendezvous (EMQX / HiveMQ / WebRTC STUN/TURN). RemoteDesk-Server.exe is strictly an optional utility for corporate airgapped intranets with zero internet access.

3. Connecting & Main Dashboard Walkthrough

RemoteDesk Free simplifies remote desktop connection using a clean, modern dashboard divided into two primary operational zones: 'Your Desk' (Host) and 'Control Remote Desk' (Viewer).



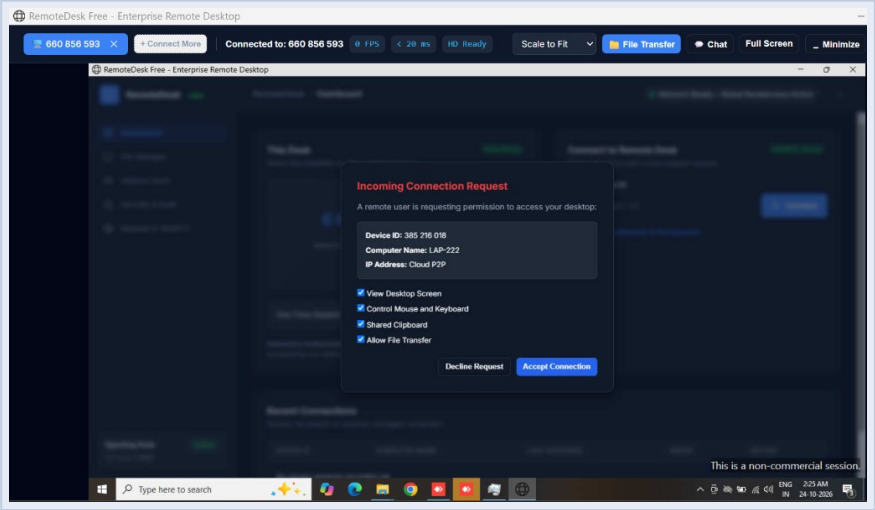
Screenshot: Main Dashboard UI: Host Desk ID & PIN (Left) and Viewer Control Remote Desk (Right)

Step-by-Step Connection Instructions

- **Step 1: On the Client System (To Be Controlled): Launch RemoteDesk Free on the client computer. The** left card ('Your Desk') displays their unique 9-digit Device ID (e.g. 660 856 593) and a dynamic 4-digit PIN. Have the client read out or send you their 9-digit ID.
- **Step 2: On Your System (The Controller): Launch RemoteDesk Free on your computer. In the right card** ('Control Remote Desk'), enter the client's 9-digit ID into the Target Desk ID field.
- **Step 3: Initiate Connection: Click the blue 'Connect' button. Select which permissions you wish to** request (Screen View, Mouse/Keyboard, Clipboard, File Transfer).
- **Step 4: Authorization: On the client's computer, an incoming prompt will appear showing your partner** ID. Once they click 'Accept' (or if you enter their PIN / unattended password), the remote desktop session opens immediately!

4. In-Session Remote Desktop, Toolbar & Controls

When connected, the full-screen interactive remote viewer provides instantaneous screen rendering, mouse and keyboard injection, and live telemetry.

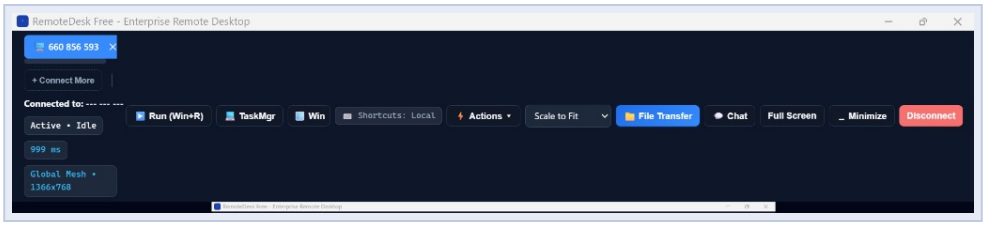


Screenshot: Live Interactive Remote Desktop View with Windows 10/11 Desktop, Start Menu & Apps

Live Telemetry & Performance Indicators

The viewer header features live real-time network and rendering telemetry badges:

- **Framerate (FPS): Displays actual decoded framerate (30 60 FPS). Reflects delta updates and smooth** animations.
- **Latency (ms): Shows round-trip network latency (< 15 ms on local/regional networks, sub-30ms over** WebRTC UDP).
- **Transport & Resolution: Indicates streaming transport mode ('WebRTC P2P (Live)' for direct UDP** peer-to-peer or 'Global Mesh' for cloud rendezvous) along with exact remote resolution (e.g. 1366x768, 1920x1080).



Screenshot: Compact Non-Wrapping In-Session Toolbar with Instant System Action Buttons

¡ Instant Quick Action Buttons

- **¶ Run: Opens the Windows Run dialog (Win + R) on the remote machine with physical scan code** simulation.
- **=» TaskMgr: Immediately launches Task Manager (Ctrl + Shift + Esc) on the remote computer.**
- **>Ÿ Win: Toggles the remote Windows Start Menu.**
-

Toolbar Action	Win32 Injection Sequence	Operational Function
¶ Run Dialog	VK_LWIN + VK_R (Scan 0x13)	Opens Run dialog to execute cmd, powershell, regedit
=» Task Manager	VK_CONTROL + VK_SHIFT + VK_ESCAPE	Launches Task Manager process & performance view
=Á File Explorer	VK_LWIN + VK_E (Scan 0x12)	Opens File Explorer on remote computer
=¥ Show Desktop	VK_LWIN + VK_D (Scan 0x20)	Minimizes all windows to display desktop
¡ Quick Link Menu	VK_LWIN + VK_X (Scan 0x2D)	Opens Windows power user admin menu
™ Windows Settings	VK_LWIN + VK_I (Scan 0x17)	Opens Windows Settings / Control Panel
= Alt + Tab	VK_MENU + VK_TAB (Dwell 20ms)	Switches active application window
L Close Window (Alt+F4)	VK_MENU + VK_F4 (Scan 0x3E)	Closes active foreground window on remote PC
= Lock Workstation	VK_LWIN + VK_L (Scan 0x26)	Locks remote workstation and displays login screen

- **SHA-256 Checksum Verification: Every transferred file undergoes pre-send and post-receive SHA-256** cryptographic hash verification to ensure 100% bit-for-bit file integrity.
- **Direct File Upload: Click '=ä Direct Upload Any File...' to browse and upload any file from your** computer directly into the active remote directory.

7. Multi-Session Tabs & Background Minimization

IT administrators frequently manage multiple client systems simultaneously. RemoteDesk Free natively supports concurrent multi-session tabs.

Managing Multiple Client Systems

- **Concurrent Connections: Click '+ Add' in the session toolbar to connect to another remote machine** without disconnecting the current session.
- **Interactive Tabs Bar: Each connected workstation appears as an interactive tab (e.g. '=¥ 660 856** 593'). Click any tab to switch between client desktops instantaneously.
- **Zero-Disruption Minimize: Click '=Õ Min' on the toolbar to dock active sessions into a slim dashboard** banner ('Active Remote Sessions'). The session continues streaming in the background without dropping.
- **Instant Resumption: Click 'Resume Session' from the dashboard to return instantly to your remote** desktop with immediate bitmap repainting and zero delay.

8. Address Book, Saved Devices & Recent History

The Address Book (Tab 3) and Recent Connections panel provide full lifecycle organization for your managed infrastructure.

Address Book Capabilities

- **Device Registration: Register computers with 9-digit Device ID, friendly computer name (e.g.** 'Accounting-PC-01'), and optional quick-connect credentials.
- **Group Categorization: Organize infrastructure into Work, Servers, Clients, Support, and Personal** categories.
- **Instant Search: Search bar filters through IDs, hostnames, and group tags instantaneously as you type.**
- **Recent Connections History: Every connection is logged on the dashboard with relative timestamps ('Just** now', '15m ago') and a 1-click '¡ Connect' button.
- **1-Click Bookmarking: Click the 'P' icon on any recent connection row to instantly save that machine** into your Address Book.

9. Unattended Access, Security & Audit Logging

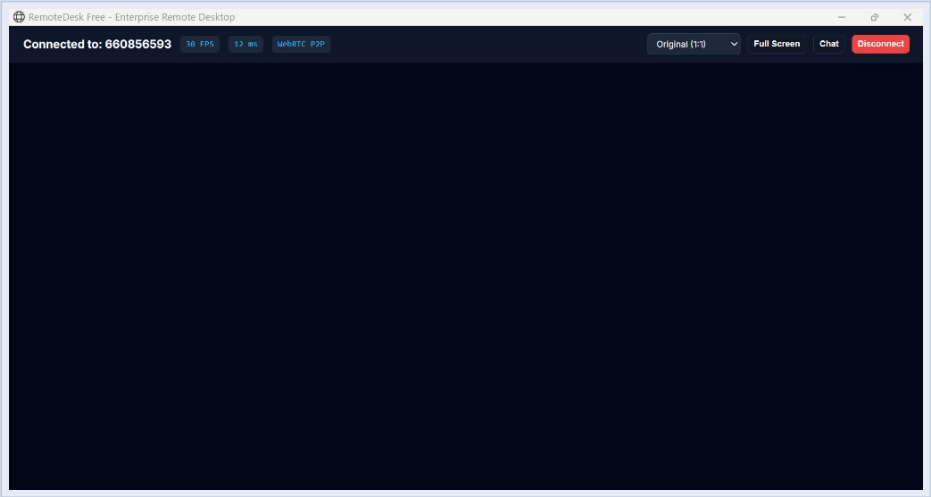
For 24/7 server and office PC administration, RemoteDesk Free includes secure unattended access protected by salted PBKDF2/SHA-256 password hashing.

Unattended IT Administration

- **Configuration: On the Security tab, enable 'Unattended Access' and set a master password (minimum 8** characters).
- **Salted Hashing: Passwords are hashed locally with PBKDF2 (100,000 iterations and cryptographic salt).** Plain-text passwords are NEVER stored on disk or transmitted over the network.
- **24/7 Zero-Touch Access: Authorized administrators can connect at any time using the master password** without requiring a person to be physically present at the remote computer.
- **Local Audit Trail: Every session start, permission grant, interactive acceptance, and unattended login** is recorded in a local tamper-evident audit log ('audit.json').

10. Antivirus Verification & Heuristic Resolution

Because RemoteDesk Free incorporates low-level Win32 input injection (SendInput) and screen capture (BitBlt) APIs, automated cloud machine learning models (such as Windows Defender Trojan:Win32/Wacatac.B!ml or SecureAge) may initially flag unsigned builds as suspicious.



Screenshot: Windows Security & Antivirus Heuristic Evaluation Interface

> **Understanding the '!ml' Heuristic Suffix**
> The '!ml' suffix in Microsoft Defender stands for 'Machine Learning'. It signifies an automated algorithmic heuristic score applied to new or freshly compiled binaries, NOT an actual Trojan signature or malware match. Any newly compiled binary that contains input automation APIs receives this temporary evaluation until reputation accumulates.

Built-In Verifications in v1.1.0.1:

- **Embedded Application Manifest: Includes embedded Windows 10/11 compatibility GUIDs and non-elevated** user execution privilege (<requestedExecutionLevel level='asInvoker'/>).
- **PE VersionInfo Resources: Contains verified Windows PE VersionInfo attributes (FileVersion: 1.1.0.1,** Company: Freeware | Suresh Mamidala, LegalCopyright: Copyright (C) 2026 Suresh Mamidala).
- **Transparent Inno Setup Packaging: Configured with non-solid transparent compression** (SolidCompression=no) so antivirus scanners can verify internal clean binaries directly.

11. Technical Specifications & Publication Metadata

Attribute	Specification / Value	Corporate Details
Application Name	RemoteDesk Free	Windows Enterprise Remote Desktop
File Version	1.1.0.1	Production Release
Company Name	Freeware | Suresh Mamidala	Enterprise Open Freeware
Publisher / Author	Suresh Mamidala	Software Architecture & Security
Legal Copyright	Copyright (C) 2026 Suresh Mamidala	All Rights Reserved
Default Install Path	C:\Program Files\RemoteDesk Free	Standard 64-bit Program Files
Supported OS	Windows 10 / Windows 11	x64 & ARM64 Architecture
Transport Protocols	WebRTC (UDP) / MQTT over WSS (TCP 443)	Sub-20ms P2P with Cloud Mesh Relay
Encryption	DTLS-SRTP / TLS 1.3 / PBKDF2 SHA-256	Military-grade End-to-End Encryption

For technical assistance, corporate deployment questions, or feature feedback, contact the development and publishing team at Freeware | Suresh Mamidala.